DATA GOVERNANCE, COMPLIANCE AND ETHICS

WEEK 01: DATA GOVERNANCE AND MANAGEMENT (PART 1)

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OVERVIEW OF COURSE CONTENT

1. Data Governance and Management (Weeks 1-2)
2. Privacy and Data Protection (Weeks 3-5)
3. Ethical Issues Pertaining to Data (Weeks 6, 8)
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 9-12)

* Week 7 = Reading Week (No classes)

AGENDA

Week 1: Introduction to Data Governance (DG)

- Definitions of DG
- Data Governance & Data Management
- Good and Poor DG
- Introduction to DMBOK wheel

Week 2 : Application of concept

- Data Management principles
- Management of data strategies, policies, processes and procedures
- Data Quality, Data Provenance, Data Integrity
- Roles and responsibilities

Supplementary Material

- Data risk identification and management
- Models and tools for data governance

DATA GROWTH

The 9th edition illustrates the impact of the coronavirus pandemic on:

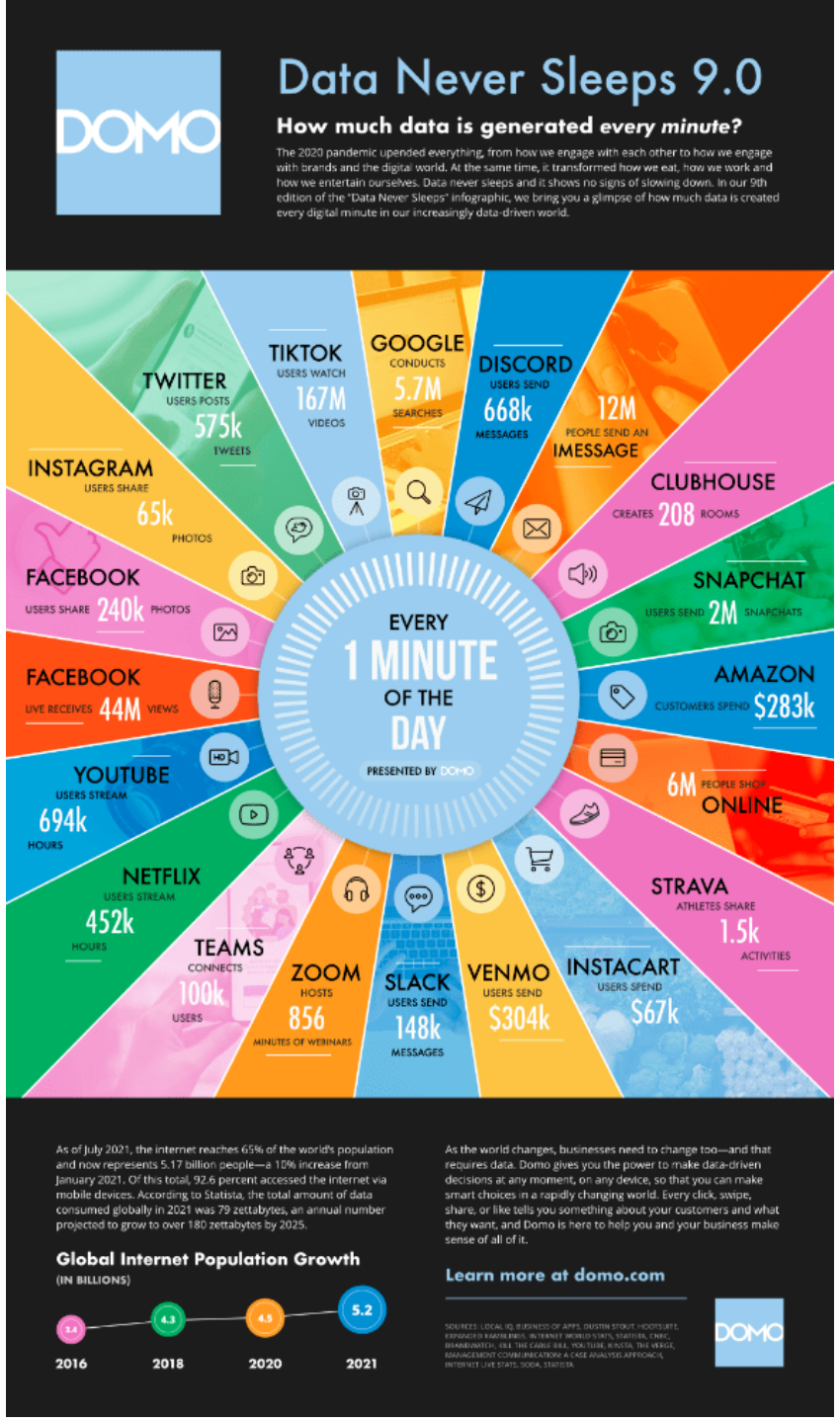
- increasing digitization of daily life
- how technology is helping re-imagine the future of work

Every minute (2021)

167M videos TikTok
>100K users connected by Microsoft
452k hours of video streamed by Netflix users

World's data doubles every 18 months *(but how much of it is useful?)*

Source: https://www.domo.com/learn/infographic/data-never-sleeps-9



DATA, AN ORGANIZATIONAL ASSET

- Data is a vital enterprise asset
- An asset is an economic resource, that can be owned or controlled, and that holds or produces value → $
- Only few organizations actively manage data as an asset
- Deriving value requires intention, planning, coordination, and commitment



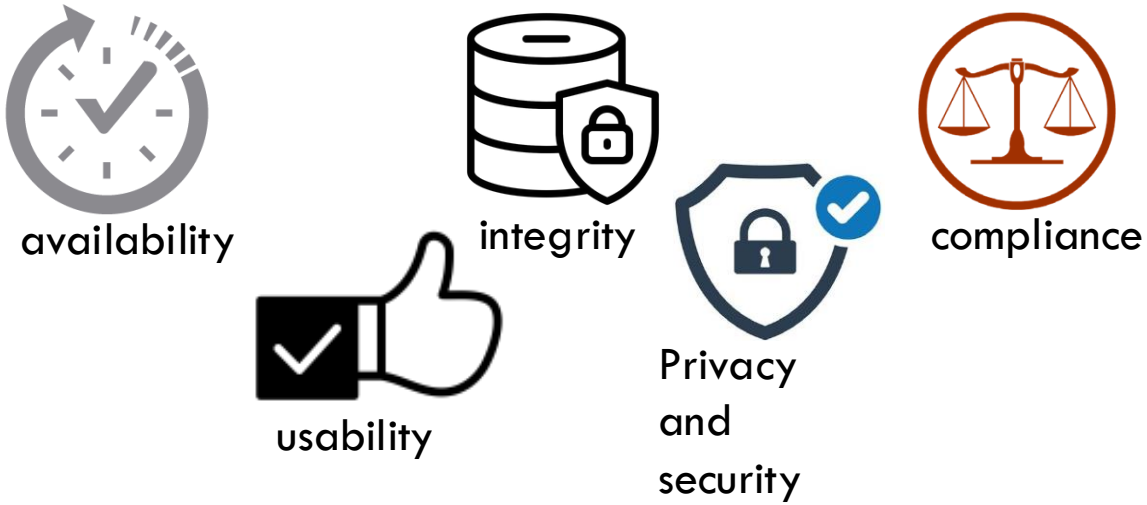


WHAT IS DATA GOVERNANCE?

- Data Governance (DG) is a collection of practices and processes which help to ensure the formal management of data assets within an organization.

(Dataversity)

- DG seeks to ensure:





- Based on:

  Internal data standards and policies

WHAT IS DATA GOVERNANCE?

Other Definitions of Data Governance Include:

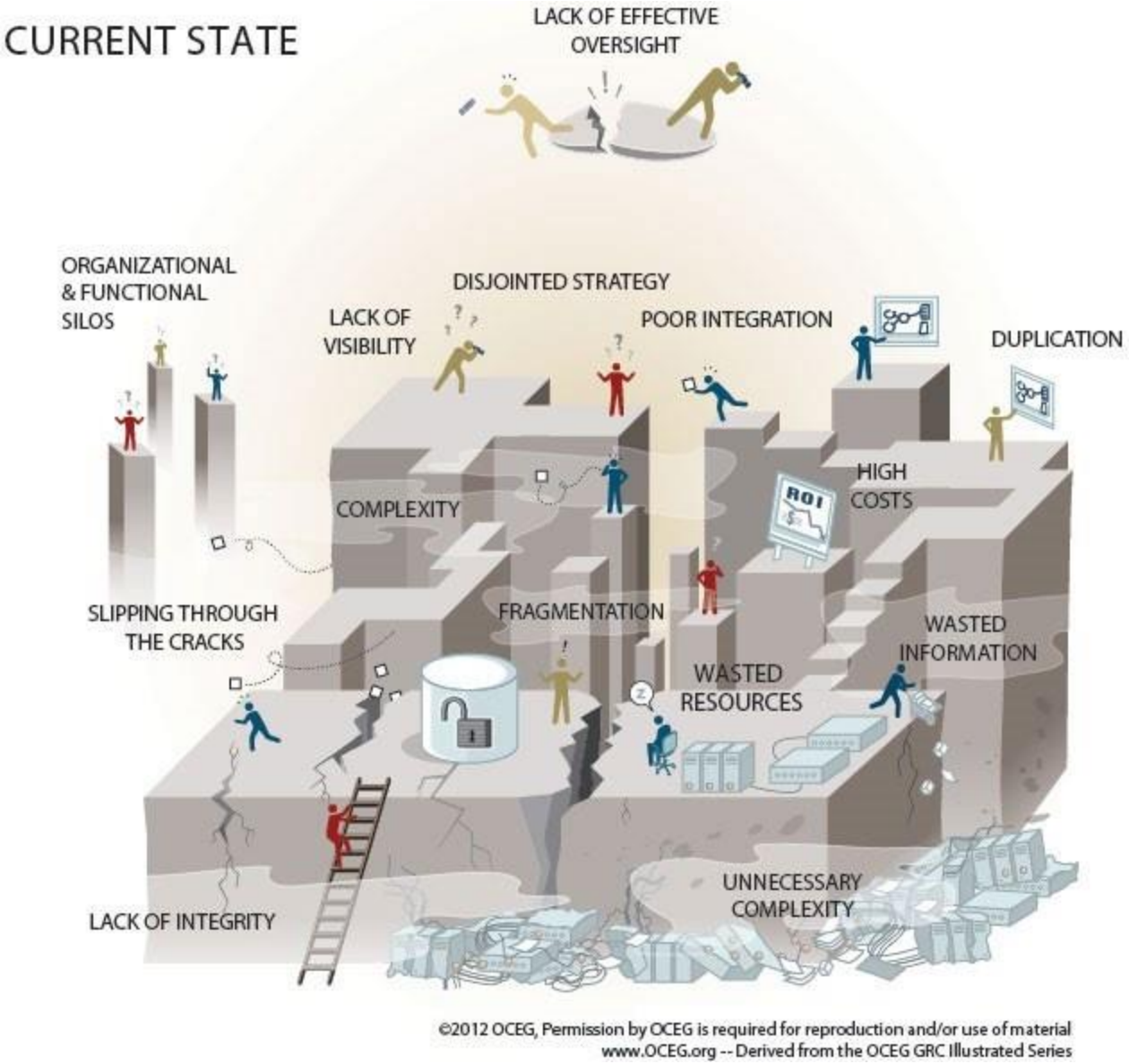
- "*Data Governance* is **the exercise of authority and control** (planning, monitoring, and enforcement) over the management of data assets." (DAMA DMBOK)

- "*Data Governance* is a system of **decision rights and accountabilities** for information-related processes, executed **according to agreed-upon models** which describe **who** can take **what actions** with **what** information, and **when**, under **what circumstances,** using **what methods.**" (Data Governance Institute)

- "*Data governance* is the organization and implementation of **policies, procedures, structure, roles, and responsibilities** which outline and enforce **rules of engagement, decision rights,** and **accountabilities** for the effective **management of information assets**" (Ladley, J., McGilvray, D., Smith, A.-M., & Thomas, G. Common)

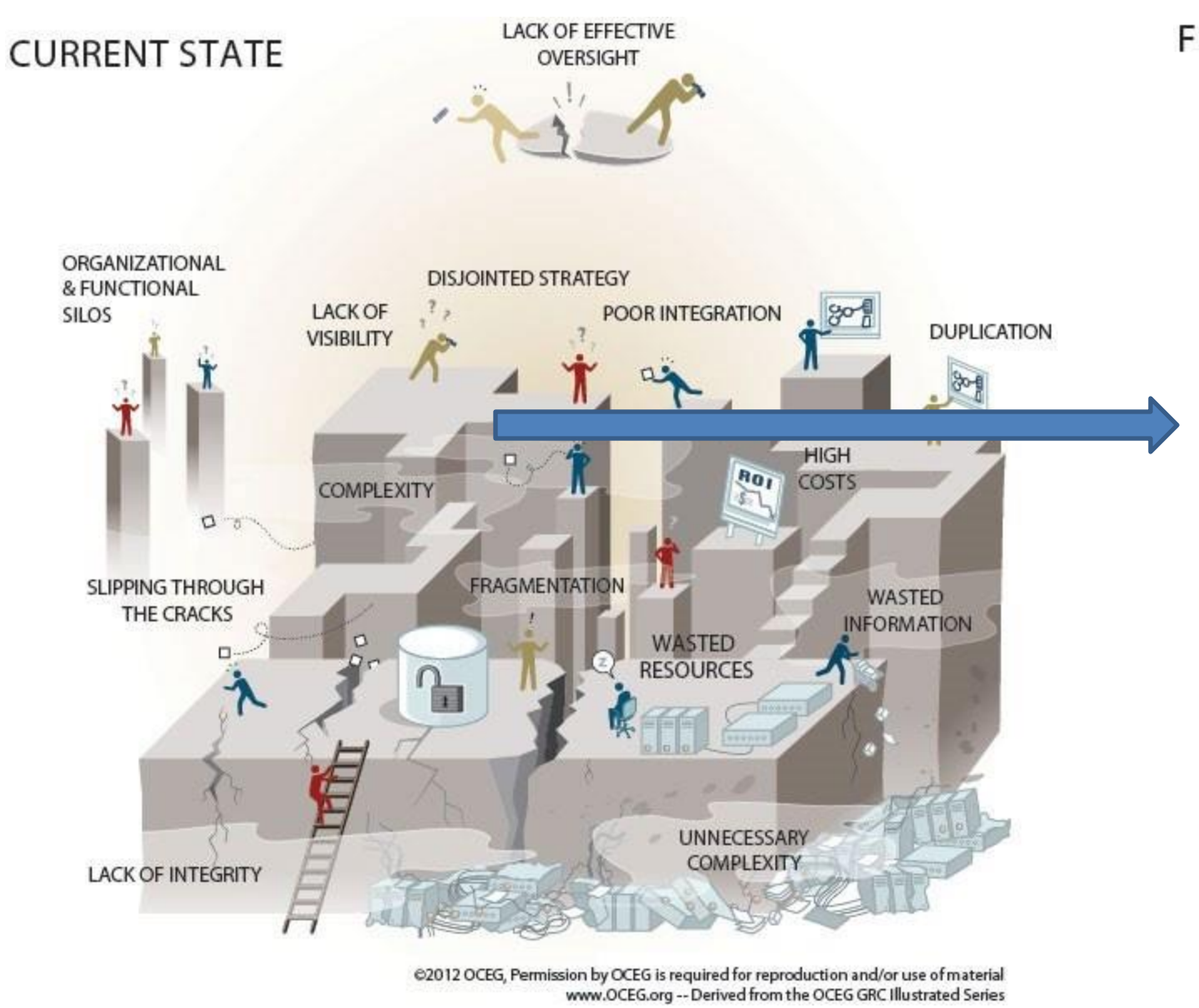
- "*Data Governance* is a discipline that provides clear-cut **policies; procedures; standards; roles; responsibilities;** and **accountabilities** to **ensure that data is well-managed** as an enterprise resource." (DGPO)

DATA GOVERNANCE - DONE WRONG

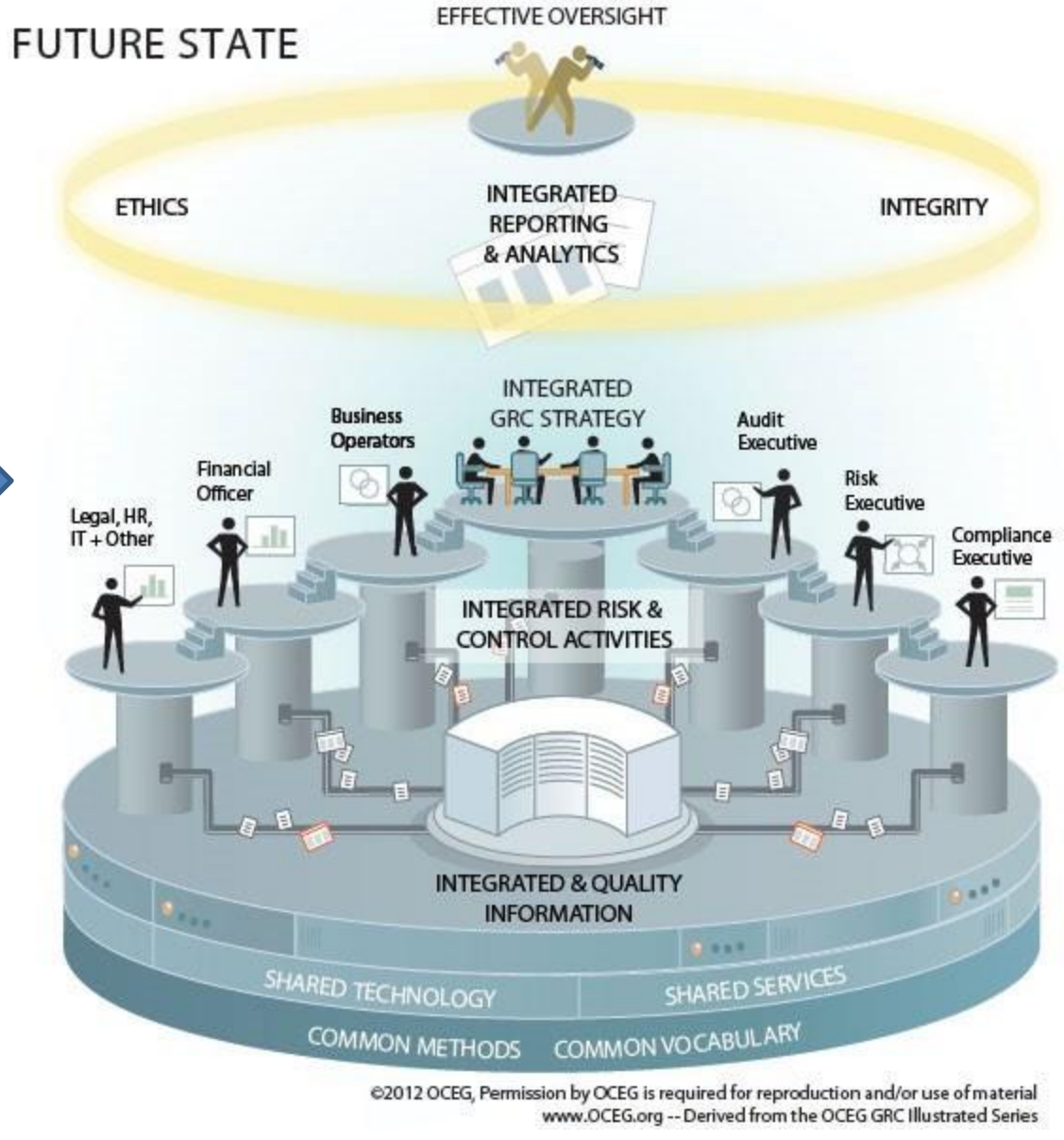


- No governance or disjointed governance activities cause a number of problems.
- Departments are often siloed
- Sub-optimal strategies are selected, or counter-productive objectives are established.

DATA GOVERNANCE - DONE RIGHT



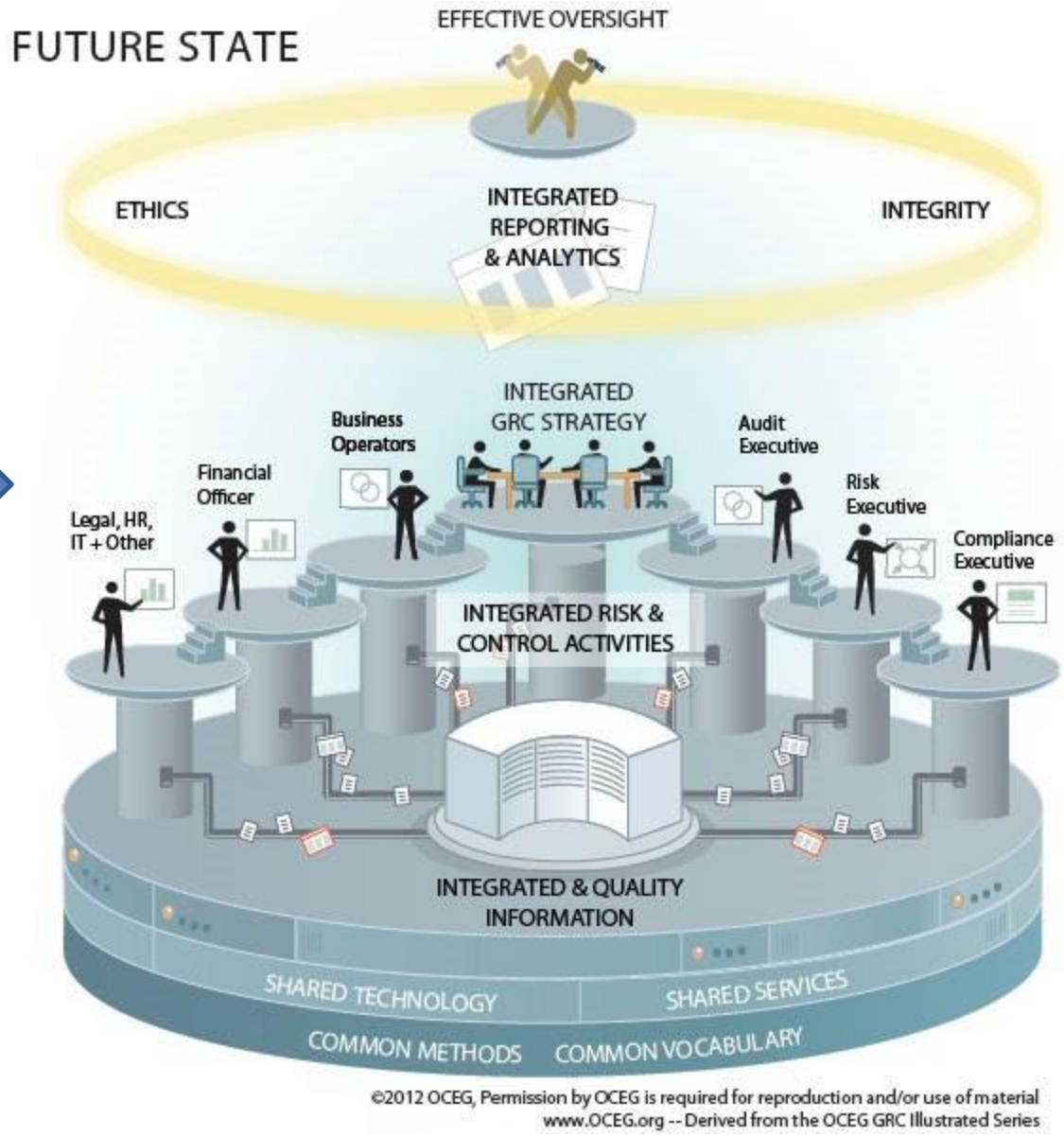
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DATA GOVERNANCE - DONE RIGHT





- ✓ Integrating Governance, Risk, & Compliance (GRC) capabilities
- ✓ Establishing an approach that ensures: the right people get the right information at the right times; the right objectives are established; the right actions and controls are put in place to address uncertainty and act with integrity
- ✓ Benefits: reduced costs, reduced duplication of activities/data, greater information quality, quick access to information, replicability.

WHY DATA GOVERNANCE MATTERS?

A good and effective data governance seeks to

- ✓ Minimize data-related risks
- ✓ Establish internal rules for responsible and ethical data use
- ✓ Implement compliance requirements
- ✓ Improve corporate communication strategy
- ✓ Increase the business value of data
- ✓ Reduce costs (business, operational, admin)
- ✓ Help to ensure the continued existence of the company through risk management and optimization















CONSEQUENCES OF POOR DATA GOVERNANCE



- Data breaches



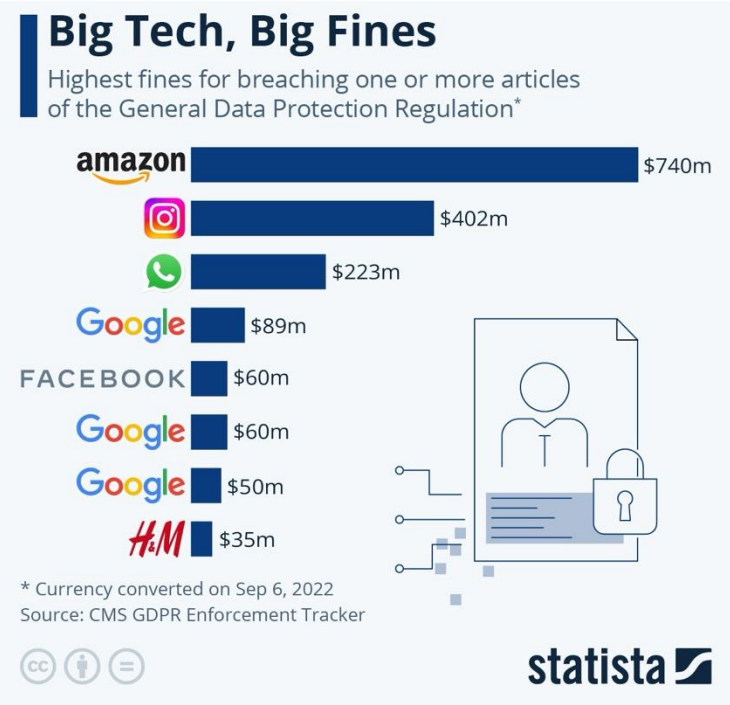
FOX 2 Detroit



EdgyLabs.com

CONSEQUENCES OF POOR DATA GOVERNANCE

- Data breaches
- Non-compliance with laws and regulations





CONSEQUENCES OF POOR DATA GOVERNANCE

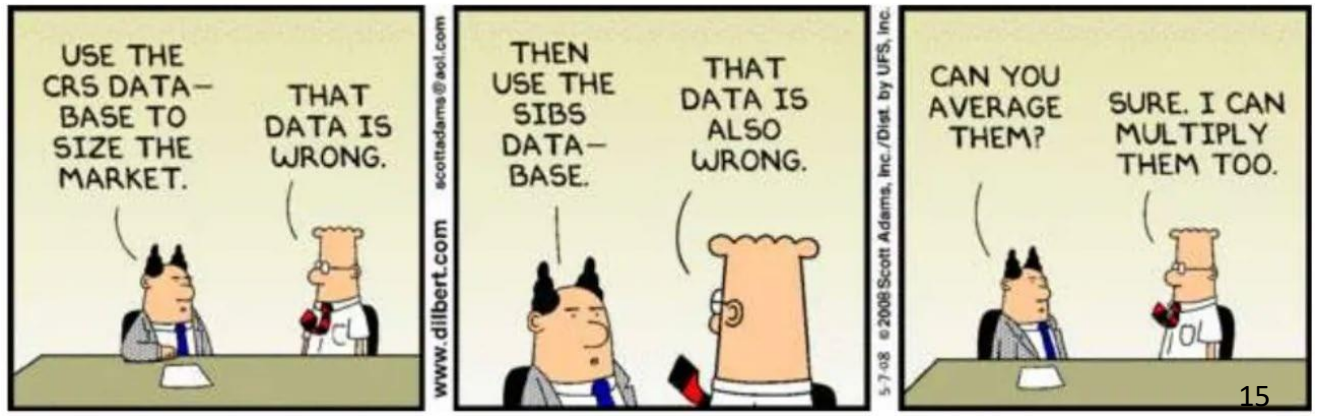
- Data breaches
- Non-compliance with laws and regulations
- Loss of reputation, customer loyalty, and revenue



https://www.datafold.com/blog/enterprises-whose-bad-data-cost-them-millions-lessons-from-samsung-and-uber

CONSEQUENCES OF POOR DATA GOVERNANCE

- Data breaches
- Non-compliance with laws and regulations
- Loss of reputation, customer loyalty, and revenue
- Poor accuracy of analytics



EXAMPLE OF POOR DATA GOVERNANCE

Former Cambridge Analytica chief receives seven-year directorship ban

Alexander Nix handed penalty for 'potentially unethical' behaviour linked to scandal



▲ Cambridge Analytica's former chief executive Alexander Nix has been barred from acting as a company director for seven years. Photograph: Tolga Akmen/AFP/Getty Images

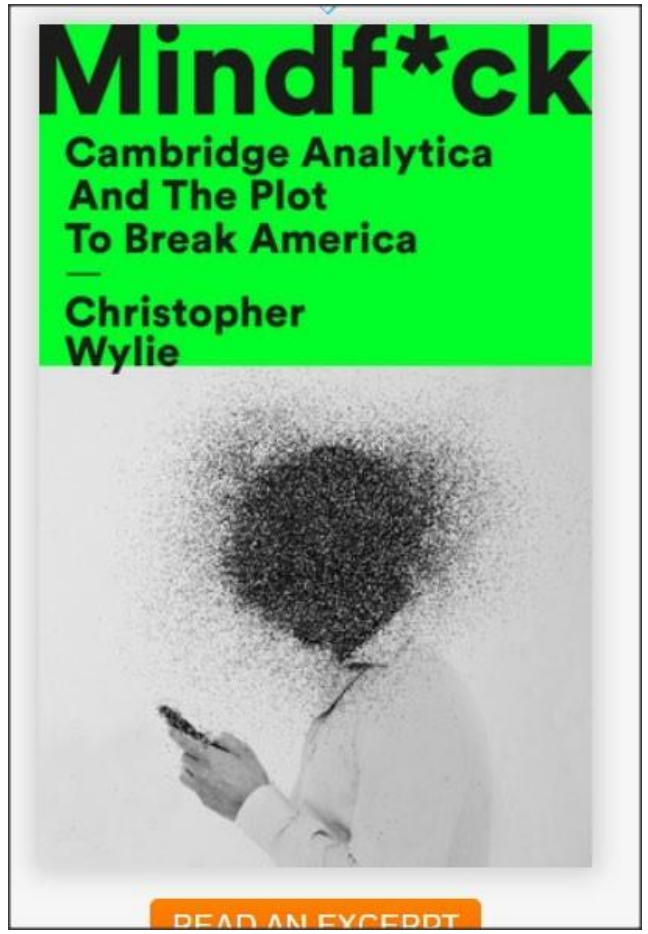
Alexander Nix, the former boss of Cambridge Analytica, has been banned from serving as a company director for seven years over "potentially unethical" behaviour linked to his position at the centre of a global scandal.

https://www.theguardian.com/uk-news/2020/sep/24/cambridge-analytica-directorship-ban-alexander-nix?CMP=Share_AndroidApp_Other





Photo by: Ledger Insights



https://www.penguinrandomhouse.com/books/604375/mindfck-by-christopher-wylie/

EXAMPLE OF POOR DATA GOVERNANCE



Facebook

Facebook-Cambridge Analytica data breach lawsuit ends in 11th hour settlement

Dramatic move shows Mark Zuckerberg 'desperate to avoid being questioned over cover-up', says Observer journalist who exposed scandal

Facebook agrees to settle Cambridge Analytica data privacy lawsuit

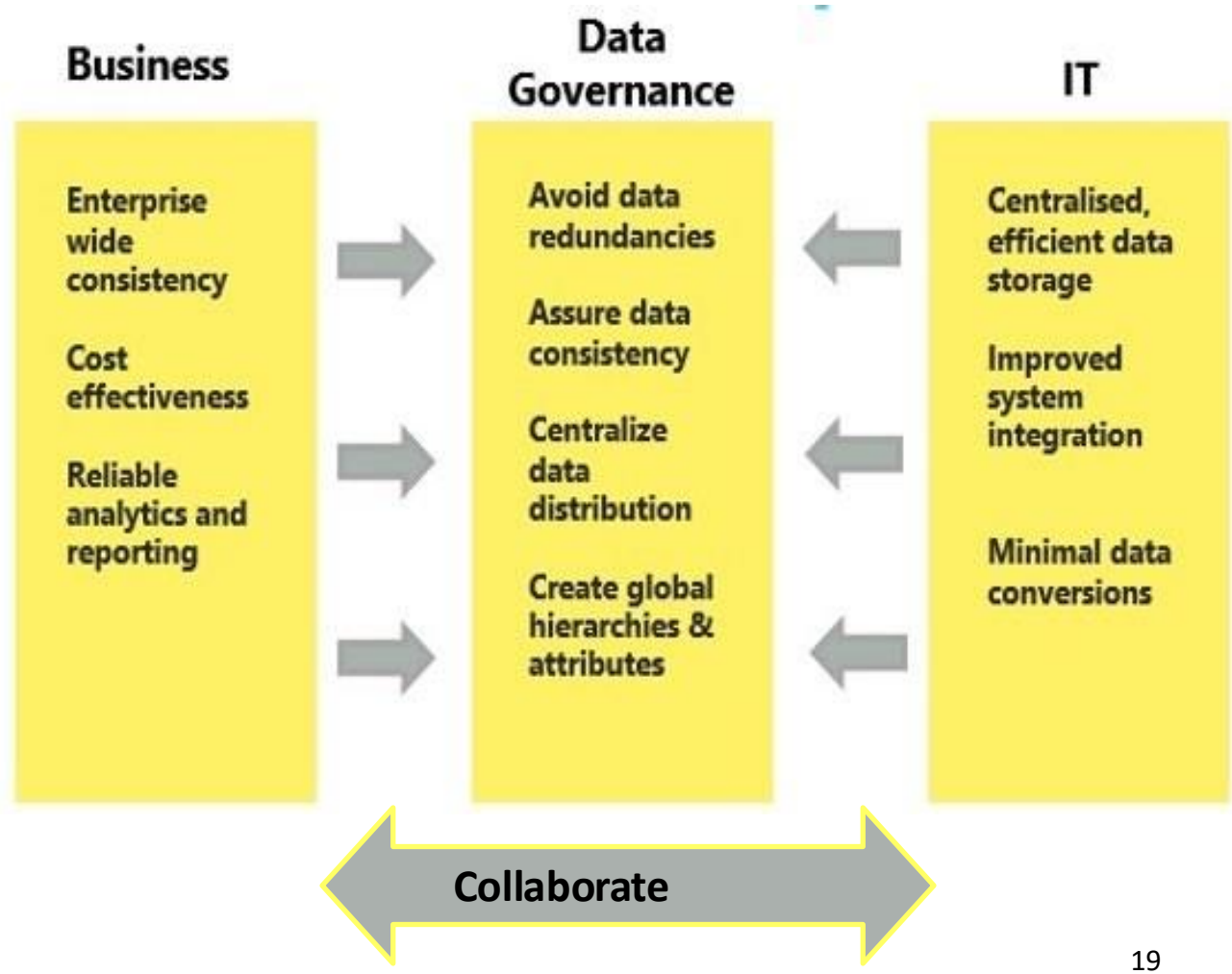
The four-year-old case alleged that the company had violated consumer privacy laws by sharing users' personal data with third parties



Facebook has agreed to settle a lawsuit that accused the company of allowing third parties to access users' private data. Photograph: Daniel Felipe Kutepov/Sopa Images/Rex/Shutterstock

DATA GOVERNANCE AS PARTNERSHIP

- DG requires a partnership between business and IT to:
  - Avoid retaining or gathering redundant data
  - Providing quality assurance about data
  - Provide an effective and consistent means of distributing data to those who need it for business purposes
  - Calibrate and empower access to appropriate use of data
- **DG should be business led with IT input not the other way around**



high quality data that meets strategic needs

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DATA GOVERNANCE: HOW?

Data Governance includes

- the people, processes and technologies needed to
- manage and protect the company's data assets
- in order to guarantee generally understandable, correct, complete, trustworthy, secure and discoverable corporate data.



DG trifecta or the PPT methodology

DATA GOVERNANCE VS DATA MANAGEMENT (1/3)

Data Management definitions:

- "A comprehensive collection of **practices, concepts, procedures, processes, and** a wide range of accompanying **systems** that allow for an organization to **gain control of its data resources.**" (Dataversity)

- "The **development, execution, and supervision** of plans, policies, programs, and practices that deliver, control, protect, and enhance the value of data and information assets throughout their lifecycles." (DAMA)

Definition of rules, processes, accountability



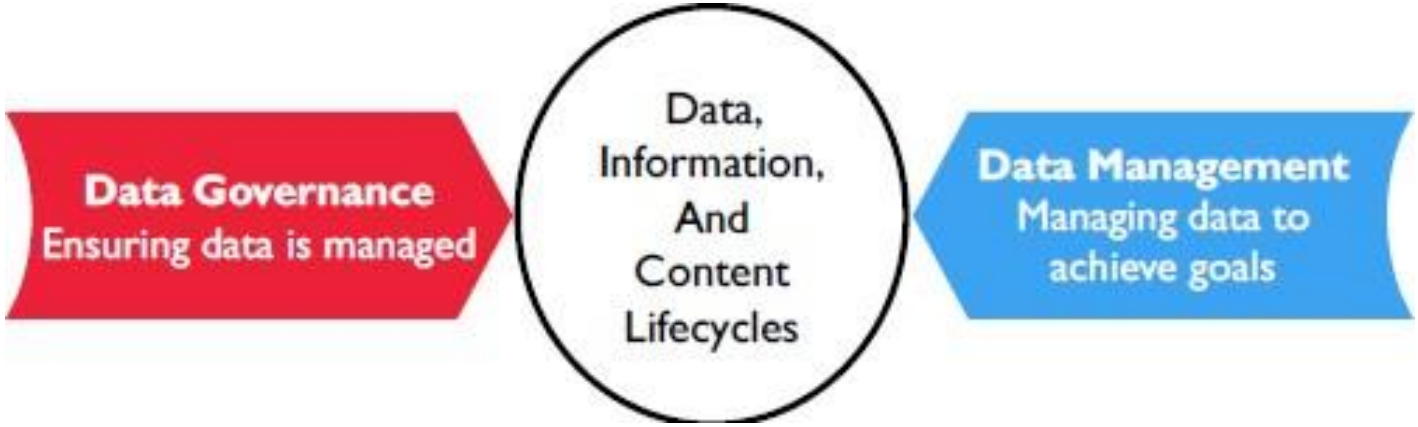
Execution/implementation of the processes.
It leverages the solutions created in DG to achieve tasks

- Example: a retention policy



DATA GOVERNANCE VS DATA MANAGEMENT (2/3)

- **Data Management** is **the process of** gathering, storing, organizing and maintaining data created and collected by an organization.
- **Data Governance** determines the **overall practice** of data management. It **is regarded as the "management of data management"** and envisages a separation between those governing and those managing

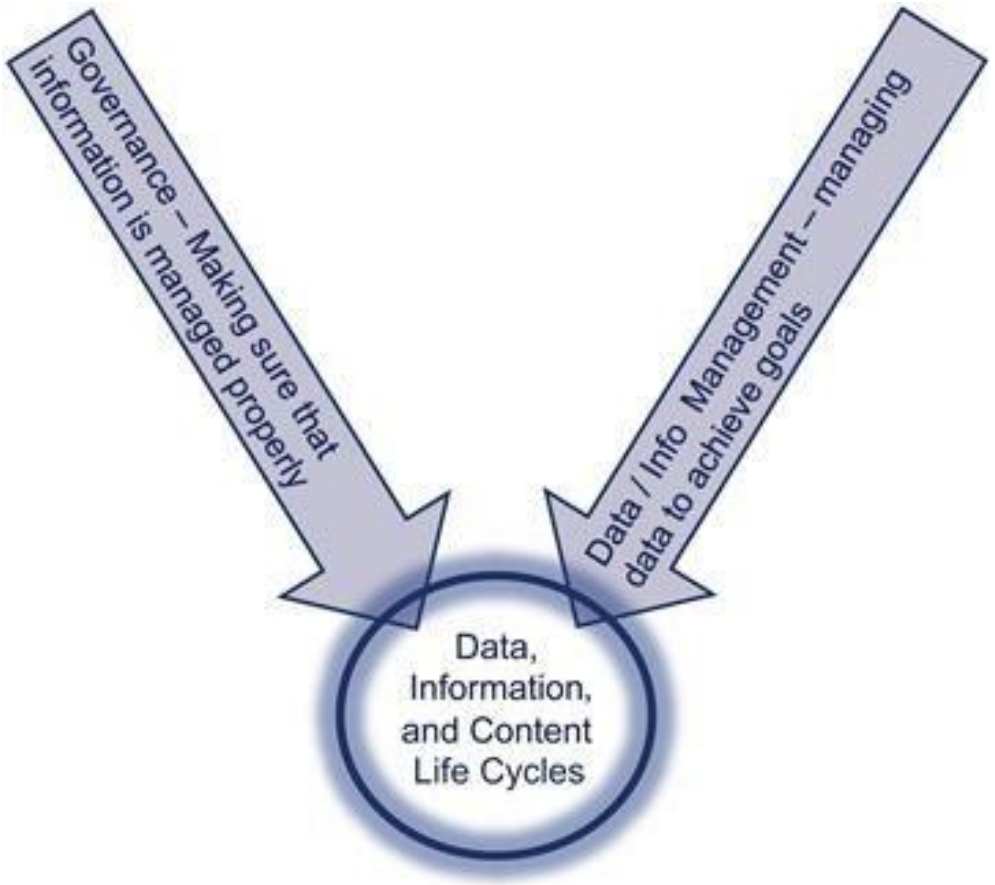


***Analogy**: auditor controls financial processes but does not actually execute financial management*

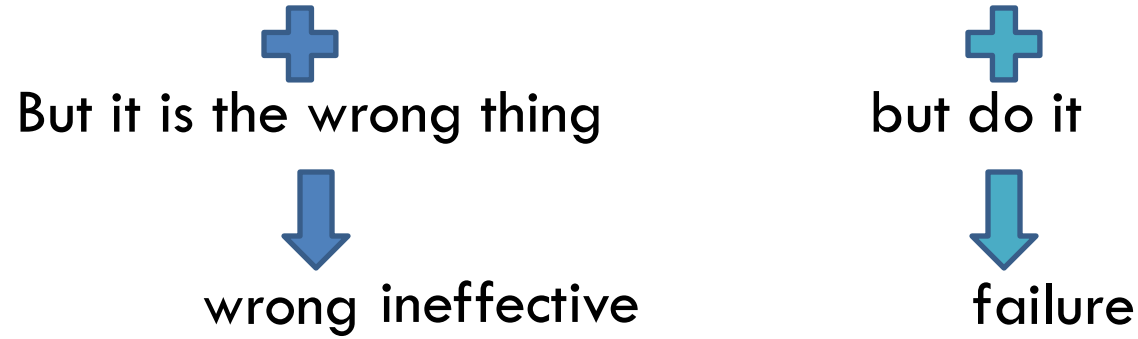


Figure 15 Data Governance and Data Management

DATA GOVERNANCE VS DATA MANAGEMENT (3/3)



"Doing things right(DM)" vs "Doing the right thing (DG)"



=> even if an organization has no specific DG practices, things still get done

=> a clear structure and roles need to defined through agreed policies, operational processes and procedure

DATA MANAGEMENT

- Data management (DM) involves a set of interdependent functions, each with its own goals, activities, and responsibilities.
- A framework helps to understand the DM comprehensively and see relationships between its component pieces.



DAMA-DMBOK2 DATA MANAGEMENT FRAMEWORK

- DM involves a set of interdependent functions
- A framework helps to understand the DM comprehensively
- An organization's approach to DG and DM depends on key factors: industry, data it uses, culture, maturity level, strategy, challenges, etc.

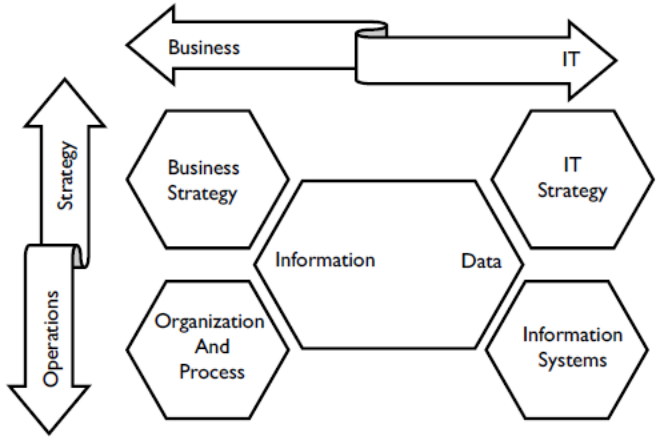


Figure 3 Strategic Alignment Model[12]

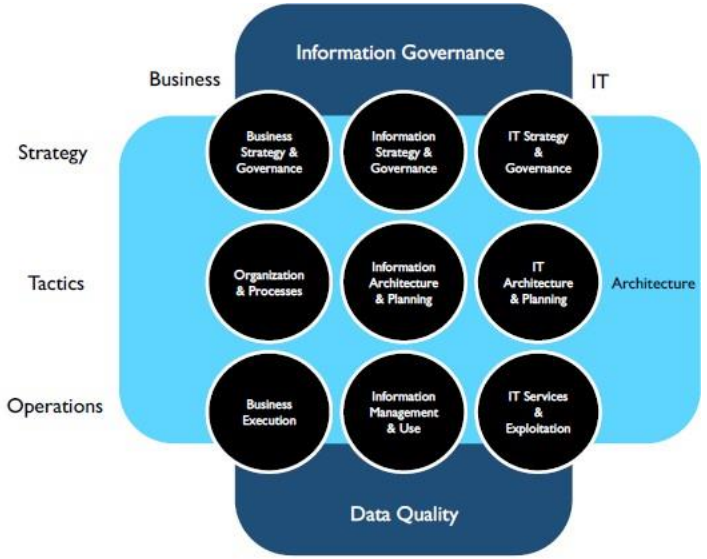


Figure 4 Amsterdam Information Model[14]

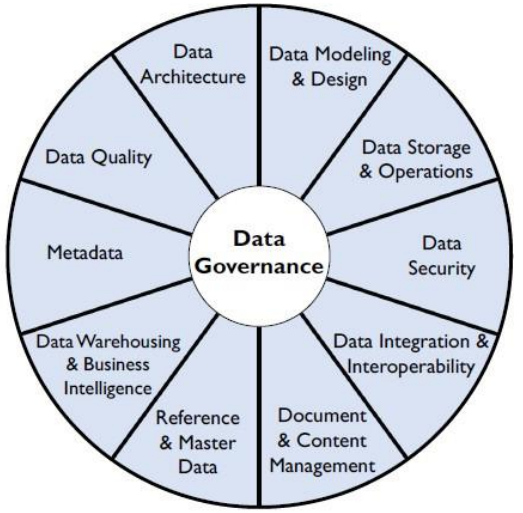
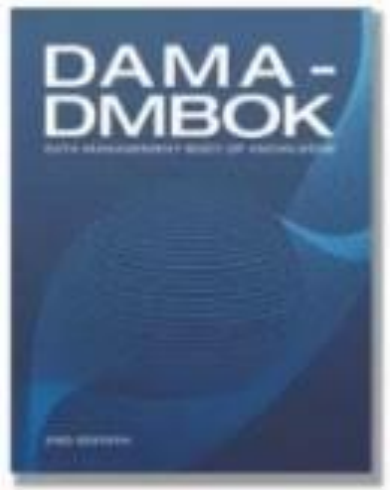
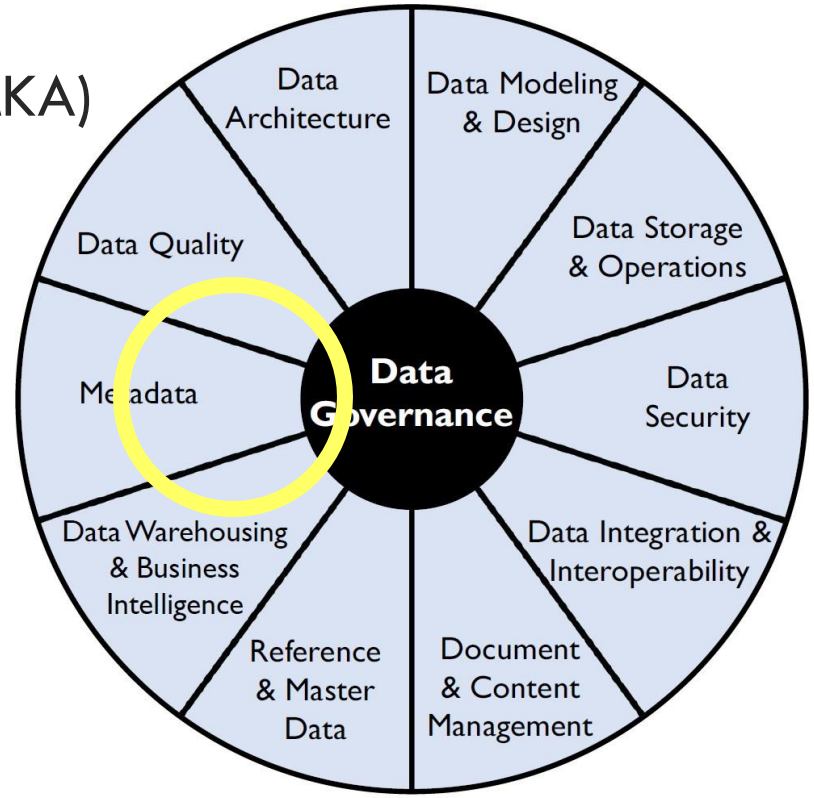


Figure 5 The DAMA-DMBOK2 Data Management Framework (The DAMA Wheel)

DAMA-DMBOK2 DATA MANAGEMENT FRAMEWORK

- DM involves a set of interdependent functions
- A framework helps to understand the DM comprehensively
- An organization's approach to DG and DM depends on key factors
- **The DAMA Wheel**
  - defines the Data Management Knowledge Areas (DMKA)
  - DMKA may be implemented at different times





DAMA-DMBOK2 Data Management Framework

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DAMA-DMBOK2 DATA GOVERNANCE

- DG focuses on how decisions are made about data and how people and processes are expected to behave in relation to data.
- The scope and focus of a particular data governance program will depend on organizational needs, but most programs include:
  - ✓ **Strategy:** Defining, communicating, and driving execution of Data Strategy and Data Governance Strategy
  - ✓ **Policy:** Setting and enforcing policies related to data and Metadata management, access, usage, security, and quality
  - ✓ **Standards and quality:** Setting and enforcing Data Quality and Data Architecture standards
  - ✓ **Oversight:** Providing hands-on observation, audit, and correction in key areas of quality, policy, and data management (often referred to as stewardship)
  - ✓ **Compliance:** Ensuring the organization can meet data-related regulatory compliance requirements
  - ✓ **Issue management:** Identifying, defining, escalating, and resolving issues related to data security, data access, data quality, regulatory compliance, data ownership, policy, standards, terminology, or data governance procedures
  - ✓ **Data management projects:** Sponsoring efforts to improve data management practices
  - ✓ **Data asset valuation:** Setting standards and processes to consistently define the business value ofdata assets

End of Week 1

QUESTIONS?